

An Elementary Proof of Fundamental Theorem of Algebra over Rationals

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Abstract

We, with the aid of AI, give an elementary proof of the Fundamental Theorem of Algebra over Rationals, i.e., for any nonconstant polynomial f with rational complex number coefficients, there exists a rational complex number z such that its quadrance $Q(f(z))$ is arbitrarily small. The proof is constructive and relies only on finite rational arithmetic, without using real numbers, infinite sets, or related concepts.

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1 Introduction

The Fundamental Theorem of Algebra is a central result in mathematics. In its classical form, it states that every nonconstant polynomial f with complex number coefficients has a complex number z such that $f(z) = 0$.

Many mathematicians contributed to the historical development of the theorem, including d'Alembert, Euler, Lagrange, Laplace, Argand, Gauss, and Cauchy [4, 7]. Of particular relevance here, Gauss attempted to prove the theorem by tracking the winding of a polynomial image around the origin, though a topological step remained to be justified [5, 1]. Cauchy later gave the winding number an integral formulation, supplying a rigorous analytic framework for the argument [3, 7].

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Smale gave a short elementary proof of the Fundamental Theorem of Algebra by using the inverse function theorem in his study of the complexity of algorithms for solving equations [8, 9]. He and Shub subsequently showed that the homotopy method combined with Newton's iteration yields an efficient algorithm with a provably good complexity bound for approximating a zero of a polynomial [10, 11, 2].

While standard proofs of the Fundamental Theorem of Algebra depend on the real numbers and infinite processes, Wildberger called for a proof grounded strictly in finite rational arithmetic and formulated two alternative versions of the theorem in his online YouTube lectures [14] in 2017. This paper directly answers his call by providing the requested proof.

Using the quadrance terminology of rational trigonometry [13], write $Q(a + ib) \equiv a^2 + b^2$. We prove the following statement:

Theorem (Main theorem). *For any nonconstant polynomial f with rational complex number coefficients and any rational number $\varepsilon > 0$, there exists a rational complex number z such that*

$$Q(f(z)) < \varepsilon^2.$$

The proof uses the winding number of finite rational closed vertex lists, rational variation bounds, and a finite triangulation to obtain a contradiction if no grid point satisfies the conclusion. It involves only finite objects, rational number computation and comparisons.

2 Notation and conventions

Throughout, the symbol $\equiv$ introduces a definition, while $=$ denotes an asserted equality. We write

$$t \in T$$

to mean that t is of type T . For example, $t \in \mathbb{Q}$ and $t \in \mathbb{Z}$ mean, respectively, that t is a rational number and that t is an integer. When no confusion can arise, we also use the ordinary notation $t \in T$ for this purpose.

We say that z is a *rational complex number*, or $z \in K$, if $z = a + ib$ for some $a, b \in \mathbb{Q}$. The notation $K^\times$ denotes the nonzero rational complex numbers. As usual, $K[X]$ denotes the polynomials in X with coefficients in K .

We identify $a + ib$ with the point $[a, b]$. For points $A_1 = [x_1, y_1]$ and $A_2 = [x_2, y_2]$, define their *quadrance* by

$$Q(A_1, A_2) \equiv (x_2 - x_1)^2 + (y_2 - y_1)^2.$$

For $u = a + ib \in K$, we use the abbreviation

$$Q(u) \equiv Q([0, 0], [a, b]) = a^2 + b^2.$$

Thus $Q(u)$ denotes the quadrance of u from the origin.

For a positive rational number R , let $\text{Sq}(R)$ denote the square with vertices

$$-R - iR, \quad R - iR, \quad R + iR, \quad -R + iR.$$

A point $z = a + ib$ lies in $\text{Sq}(R)$ precisely when $|a| \leq R$ and $|b| \leq R$, and lies on its boundary $\partial\text{Sq}(R)$ precisely when, in addition, $|a| = R$ or $|b| = R$. Here $|\cdot|$ denotes the usual absolute value on $\mathbb{Q}$.

For any three-point list (p_1, p_2, p_3) , we say that z is a *nonnegative rational affine combination* of the list precisely when

$$z = \alpha p_1 + \beta p_2 + \gamma p_3, \quad \alpha, \beta, \gamma \in \mathbb{Q}, \quad \alpha, \beta, \gamma \geq 0, \quad \alpha + \beta + \gamma = 1.$$

When p_1, p_2, p_3 are distinct and non-collinear, this condition means that z lies on the triangle $p_1 p_2 p_3$. Such a point is interior to the triangle precisely when $\alpha, \beta, \gamma > 0$, and it lies on a side precisely when $\alpha\beta\gamma = 0$. When the three-point list is degenerate, we retain the affine-combination language and do not call it a triangle.

3 Rational quadrance and variation estimates

Definition 3.1 (Auxiliary size). For $u = a + ib \in K$, define

$$\sigma(u) \equiv |a| + |b|.$$

$\sigma(u)$ is called the *size* of u . Note that $Q(u) = u \cdot \bar{u}$, where $\overline{a + ib} \equiv a - ib$.

Lemma 3.2 (Basic inequalities). For all $u, v \in K$:

$$Q(u) \leq \sigma(u)^2, \quad \sigma(u + v) \leq \sigma(u) + \sigma(v), \quad \sigma(u \cdot v) \leq \sigma(u)\sigma(v). \quad (2.1)$$

Moreover σ is nonnegative, $\sigma(0) = 0$, $\sigma(1) = 1$, and $\sigma(u^n) \leq \sigma(u)^n$ for all $n \geq 0$.

Proof. Set $u \equiv a + ib$, $v \equiv c + id$. Then

$$\sigma(u + v) = |a + c| + |b + d| \leq |a| + |c| + |b| + |d| = \sigma(u) + \sigma(v).$$

$$\sigma(uv) = |ac - bd| + |ad + bc| \leq |a||c| + |b||d| + |a||d| + |b||c| = (|a| + |b|)(|c| + |d|) = \sigma(u)\sigma(v).$$

Finally, $Q(u) = a^2 + b^2 \leq (|a| + |b|)^2 = \sigma(u)^2$. The facts $\sigma(0) = 0$, $\sigma(1) = 1$, $\sigma \geq 0$ are immediate; $\sigma(u^n) \leq \sigma(u)^n$ follows from submultiplicativity by induction. $\square$

Lemma 3.3 (Power variation). Let $w, z \in K$ and $j \geq 0$, and let $B \in \mathbb{Q}$ with $\sigma(w), \sigma(z) \leq B$. Then

$$\sigma(w^j - z^j) \leq j \cdot B^{j-1} \cdot \sigma(w - z). \quad (2.2)$$

(The exponent $j - 1$ is taken in $\mathbb{Z}$; for $j = 0$ both sides are 0.)

Proof. For $j = 0$ both sides vanish. For $j \geq 1$, we note that

$$w^j - z^j = (w - z) \cdot \sum_{r=0}^{j-1} w^{j-1-r} z^r.$$

The result follows from the triangle inequality and submultiplicativity of σ , and also the fact that $\sigma(w), \sigma(z) \leq B$. $\square$

Lemma 3.4 (Polynomial variation). Let $f(X) = \sum_{j=0}^n a_j X^j \in K[X]$ and let $B \in \mathbb{Q}$ with $\sigma(w), \sigma(z) \leq B$. Then

$$\sigma(f(w) - f(z)) \leq \sigma(w - z) \cdot \sum_{j=1}^n j \sigma(a_j) B^{j-1}. \quad (2.3)$$

Proof. By the power-variation estimate (2.2) and the submultiplicative inequality in (2.1),

$$\sigma(a_j w^j - a_j z^j) \leq \sigma(a_j) j B^{j-1} \sigma(w - z).$$

Summing this inequality over $j = 1, \dots, n$ and applying the triangle inequality in (2.1) gives (2.3). $\square$

Lemma 3.5 (Affine exclusion). Let $a, b, c \in K$ and $0 < \varepsilon \in \mathbb{Q}$. If $\sigma(a - b) < \varepsilon$, $\sigma(a - c) < \varepsilon$, and $\varepsilon^2 \leq Q(a)$, then the origin is not a nonnegative rational affine combination of (a, b, c) .

Proof. If the origin were such an affine combination, then $a = \beta(a - b) + \gamma(a - c)$ with $\beta, \gamma \geq 0$, $\beta + \gamma \leq 1$, so $\sigma(a) \leq \beta\sigma(a - b) + \gamma\sigma(a - c) < \varepsilon$, giving $Q(a) \leq \sigma(a)^2 < \varepsilon^2$, contradicting $\varepsilon^2 \leq Q(a)$. $\square$

4 A winding number for finite closed vertex lists

4.1 Definitions and the half-open convention

Throughout this section, a *closed vertex list* is an oriented list $P \equiv (p_0, p_1, \dots, p_s)$ with $p_j \in K$ and $p_s = p_0$. The closed list (a, b, c, a) is the *three-cycle* determined by (a, b, c) ; its entries need not be distinct or non-collinear.

Definition 4.1 (Determinant). For $u = x + iy$, $v = x' + iy'$, define

$$\Delta(u, v) \equiv xy' - yx'.$$

This is twice the signed area of the oriented three-point list $(0, u, v)$. The sign decisions below depend on this rational expression.

Definition 4.2 (Signed crossing, half-open convention). For a directed edge $p \rightarrow q$, we define the *signed crossing contribution* $\chi(p, q) \in \mathbb{Z}$ by the half-open rule

$$\chi(p, q) \equiv \begin{cases} 1, & \text{Im}(p) \leq 0 < \text{Im}(q) \text{ and } \Delta(p, q) > 0, \\ -1, & \text{Im}(q) \leq 0 < \text{Im}(p) \text{ and } \Delta(p, q) < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Here $\text{Im}(p)$ denotes the imaginary part of p .

Remark 4.3 (Interpretation of (1)). Let's consider the case

$$\text{Im}(p) \leq 0 < \text{Im}(q) \text{ and } \Delta(p, q) > 0,$$

Write $p = x + iy$, $q = x' + iy'$ with $\text{Im}(p) = y \leq 0$ and $\text{Im}(q) = y' > 0$, and let the intersection of pq and $\text{Im}(z) = 0$ be $r = x_r, x_r \in \mathbb{Q}$. Then $\Delta(p, q) = x_r(y' - y)$. Since $y' - y > 0$, $\Delta(p, q) > 0$ iff $\text{Re}(r) > 0$.

Definition 4.4 (Winding number). For a closed vertex list $P = (p_0, \dots, p_s)$, its *winding number* is defined as follows:

$$W(P) \equiv \sum_{i=0}^{s-1} \chi(p_i, p_{i+1}) \in \mathbb{Z}.$$

It is computable by finitely many rational number computations and comparisons.

Remark 4.5 (Positive scale invariance). Let $a \in \mathbb{Q}$ be positive and write $aP \equiv (ap_0, ap_1, \dots, ap_s)$. Then

$$W(aP) = W(P).$$

Indeed, for any edge $p_i \rightarrow p_{i+1}$,

$$\text{Im}(ap_i) = a \text{Im}(p_i), \quad \Delta(ap_i, ap_{i+1}) = a^2 \Delta(p_i, p_{i+1}).$$

Since $a > 0$, all inequalities and signs in Definition 4.2 are unchanged. Hence $\chi(ap_i, ap_{i+1}) = \chi(p_i, p_{i+1})$ for any i , and summing over the edges gives $W(aP) = W(P)$ by Definition 4.4.

Theorem 4.6 (Structural properties of W). *The winding number has the following properties, all of which can be proved by finite rational computations.*

- (1) Reversal. $\chi(q, p) = -\chi(p, q)$; hence reversing all edges of a closed vertex list negates its winding number.
- (2) Three-cycle vanishing. *If the origin is not a nonnegative rational affine combination of (a, b, c) , where $a, b, c \in K$, then $W(a, b, c, a) = 0$.*

- (3) Additivity under common triangulation. Let $P \equiv (p_0, \dots, p_n)$ and $Q \equiv (q_0, \dots, q_m)$ be closed vertex lists. Inserting repeated vertices (equivalently, inserting zero-length edges) does not change winding number, so we may assume $m = n$. For each $i = 0, \dots, n-1$, suppose the origin is not a nonnegative rational affine combination of either (p_i, p_{i+1}, q_{i+1}) or (p_i, q_{i+1}, q_i) . Then

$$W(P) = W(Q).$$

- (4) Base value. For rational $R > 0$, the counterclockwise boundary of the square $\text{Sq}(R)$ has winding number 1:

$$W(-R - iR, R - iR, R + iR, -R + iR, -R - iR) = 1.$$

Proof. (1) Exchanging p and q swaps the two branches of (1) and negates $\Delta(p, q)$, giving $\chi(q, p) = -\chi(p, q)$.

(2) Assume that the origin is not a nonnegative rational affine combination of (a, b, c) . We analyse the points r on the three directed edges of the closed list (a, b, c, a) which satisfy $\text{Im}(r) = 0$ and $\text{Re}(r) > 0$. By the affine-combination condition and the exclusion of 0, there are either no relevant crossing points, one such point, or two such points. If a whole edge has imaginary coordinate zero and positive real coordinate, we treat its two endpoints as the relevant points in Case 3 below.

Case 1: no point r on an edge satisfies $\text{Im}(r) = 0$ and $\text{Re}(r) > 0$. Then no edge has such a crossing point, so each of the three contributions is 0, and the total is 0.

Case 2: exactly one point v on an edge satisfies $\text{Im}(v) = 0$ and $\text{Re}(v) > 0$. Then v must be one of a, b, c , and the imaginary coordinates of the other two vertices have the same sign. If they are positive, the two edges incident to v are one upward edge (+1) and one downward edge (-1), while the opposite edge has no crossing; total 0. If they are negative, both incident edges contribute 0, again total 0.

Case 3: the two relevant points lie on the same edge. Then that edge has zero imaginary coordinate and contributes 0. If the third vertex has positive imaginary coordinate, the other two edges give one upward crossing (+1) and one downward crossing (-1); if it has negative imaginary coordinate, both give 0. In either subcase the total is 0.

Case 4: the two relevant points are interior points of two different edges. Then those two edges cross the line $\text{Im}(z) = 0$ at points with positive real coordinate, in opposite directions: one upward (+1) and one downward (-1). The third edge has no intersection point with positive real coordinate and contributes 0. Hence the total is 0.

Case 5: one relevant point is a vertex v satisfying $\text{Im}(v) = 0$ and $\text{Re}(v) > 0$, and the other relevant point is interior to the opposite edge. Let the remaining vertices be u and w , with $\text{Im}(u) > 0$ and $\text{Im}(w) < 0$. After possibly reversing the cyclic orientation of the three-cycle, suppose the opposite edge is oriented from w to u , so it is an upward crossing and contributes +1. Then the edge from u to v is downward and contributes -1, while the edge from v to w contributes 0. Thus the total is $+1 - 1 + 0 = 0$. If the opposite edge is oriented from u to w , all signs are reversed, and the total remains 0.

In every possible case,

$$\chi(a, b) + \chi(b, c) + \chi(c, a) = 0.$$

Therefore $W(a, b, c, a) = 0$.

- (3) For each i , apply the three-cycle vanishing property established in part (2) above to (p_i, p_{i+1}, q_{i+1}) and (p_i, q_{i+1}, q_i) , obtaining

$$W(p_i, p_{i+1}, q_{i+1}, p_i) = 0, \quad W(p_i, q_{i+1}, q_i, p_i) = 0.$$

Summing these $2n$ equalities, each interior edge appears twice with opposite orientations and therefore cancels. Only the boundary edges of P and Q remain, yielding $W(P) = W(Q)$.

(4) Direct computation: among the four directed edges, only the right side $R - iR \rightarrow R + iR$ has an upward crossing point r with $\text{Im}(r) = 0$ and $\text{Re}(r) > 0$, and $\Delta(R - iR, R + iR) = 2R^2 > 0$, contributing $+1$; the other three edges contribute 0 . $\square$

5 The winding number of the power cycle

Let R be a positive rational number, and let $M \geq 2$ be an even integer. Set $h \equiv 2R/M$ and subdivide the counterclockwise boundary of $\text{Sq}(R)$ into $4M$ segments, each of quadrance h^2 . Let $z_0 \equiv R$, let $z_1, \dots, z_{4M-1}$ be the remaining successive vertices in counterclockwise order, and set $z_{4M} \equiv z_0$.

Definition 5.1. (Power cycle): The *power cycle* of degree n for $(z_0, z_1, \dots, z_{4M})$ is $P_n \equiv (z_0^n, z_1^n, \dots, z_{4M}^n)$.

For a given power cycle $P_n \equiv (z_0^n, z_1^n, \dots, z_{4M}^n)$, let $I_n \equiv W(P_n)$ denote the *winding number* of P_n and let $E_n \equiv n(2R)^{n-1}h$.

Lemma 5.2 (Parametric bounds). *For any vertex z_j ($0 \leq j \leq 4M$):*

$$Q(z_j) \geq R^2, \quad \sigma(z_j) \leq 2R, \quad \sigma(z_{j+1} - z_j) = h \quad (0 \leq j < 4M).$$

Consequently, for each $0 \leq j < 4M$,

$$\sigma(z_{j+1}^n - z_j^n) \leq E_n.$$

Proof. Since each z_j lies on $\partial\text{Sq}(R)$, $Q(z_j) \geq R^2$ and $\sigma(z_j) \leq 2R$. The difference $z_{j+1} - z_j$ is one of $\pm h, \pm hi$ so $\sigma(z_{j+1} - z_j) = h$. The last bound comes from Lemma 3.3 with $B = 2R$: $\sigma(z_{j+1}^n - z_j^n) \leq n(2R)^{n-1}h = E_n$. $\square$

Lemma 5.3 (Base winding number). $I_1 = W(z_0, \dots, z_{4M}) = 1$.

Proof. The edges on bottom, top, and left sides contribute 0 . On the right side, exactly two edges have the common endpoint R , which satisfies $\text{Im}(R) = 0$ and $\text{Re}(R) = R > 0$. According to the half-open convention, the edge leaving R upward contributes $+1$, the other one contributes 0 . Every other segment contributes 0 . Hence the total is 1 . $\square$

Theorem 5.4 (Additivity of the winding number). *Let $r, s \geq 1$. Assume the Fineness Condition $4E_{r+s}^2 < R^{2(r+s)}$. Then*

$$I_{r+s} = I_r + I_s. \quad (4.1)$$

Proof. On the parameter triangle $0 \leq v \leq u \leq 1$, consider the map $H(u, v) \equiv \Gamma_R(u)^r \Gamma_R(v)^s$. Here Γ_R is the counterclockwise rational piecewise-affine parametrization of $\partial\text{Sq}(R)$ starting at R , given explicitly by

$$\Gamma_R(t) \equiv \begin{cases} R + 8Rt i, & 0 \leq t \leq \frac{1}{8}, \\ (2R - 8Rt) + Ri, & \frac{1}{8} < t \leq \frac{3}{8}, \\ -R + (4R - 8Rt)i, & \frac{3}{8} < t \leq \frac{5}{8}, \\ (-6R + 8Rt) - Ri, & \frac{5}{8} < t \leq \frac{7}{8}, \\ R + (8Rt - 8R)i, & \frac{7}{8} < t \leq 1. \end{cases}$$

The first and last branches together in the above definition of Γ_R parametrize the right side of the square; the other three branches parametrize the top, left, and bottom sides, respectively.

Set $N \equiv 4M$ and consider the rational parameter grid

$$\mathcal{D}_N \equiv \left\{ \left(\frac{i}{N}, \frac{j}{N} \right) : 0 \leq j \leq i \leq N \right\}$$

in the triangle $0 \leq v \leq u \leq 1$. Triangulate it by the oriented three-point lists

$$\begin{aligned} T_{i,j}^+ &\equiv \left(\left(\frac{i}{N}, \frac{j}{N} \right), \left(\frac{i+1}{N}, \frac{j}{N} \right), \left(\frac{i+1}{N}, \frac{j+1}{N} \right) \right), \quad 0 \leq j \leq i < N, \\ T_{i,j}^- &\equiv \left(\left(\frac{i}{N}, \frac{j}{N} \right), \left(\frac{i+1}{N}, \frac{j+1}{N} \right), \left(\frac{i}{N}, \frac{j+1}{N} \right) \right), \quad 0 \leq j < i < N. \end{aligned}$$

For any such elementary triangle $T = (A, B, C)$, its *image three-point list* is $(H(A), H(B), H(C))$, and its *image three-cycle* is $(H(A), H(B), H(C), H(A))$.

Under H , the side $v = 0$ maps to $z_0^s \cdot P_r$, the side $u = 1$ maps to $z_0^r \cdot P_s$, and the diagonal $u = v$ maps to P_{r+s} . For every $(u, v) \in \mathcal{D}_N$, both $\Gamma_R(u)$ and $\Gamma_R(v)$ lie on $\partial \text{Sq}(R)$. Hence

$$\sigma(\Gamma_R(u)), \sigma(\Gamma_R(v)) \leq 2R, \quad Q(\Gamma_R(u)), Q(\Gamma_R(v)) \geq R^2,$$

and therefore, by the multiplicativity of quadrance,

$$Q(H(u, v)) = Q(\Gamma_R(u))^r Q(\Gamma_R(v))^s \geq R^{2(r+s)}. \quad (4.2)$$

Let (u, v) and (u', v') be any two vertices of one elementary triangle, and write

$$x \equiv \Gamma_R(u), \quad x' \equiv \Gamma_R(u'), \quad y \equiv \Gamma_R(v), \quad y' \equiv \Gamma_R(v').$$

The two parameter coordinates differ by at most one grid step, so the corresponding boundary vertices are either equal or consecutive. Thus

$$\sigma(x' - x) \leq h, \quad \sigma(y' - y) \leq h.$$

Applying Lemma 3.4 to the monomials X^r and X^s gives

$$\sigma(x'^r - x^r) \leq r(2R)^{r-1}h, \quad \sigma(y'^s - y^s) \leq s(2R)^{s-1}h.$$

Using

$$H(u', v') - H(u, v) = (x'^r - x^r)y'^s + x^r(y'^s - y^s)$$

together with the triangle and submultiplicative inequalities in Lemma 3.2, we obtain

$$\begin{aligned} \sigma(H(u', v') - H(u, v)) &\leq r(2R)^{r-1}h(2R)^s + (2R)^r s(2R)^{s-1}h \\ &= (r+s)(2R)^{r+s-1}h = E_{r+s}. \end{aligned}$$

Now let (a, b, c) be the image three-point list of any elementary triangle. Applying the preceding estimate to its first vertex and each of the other two gives

$$\sigma(a - b) \leq E_{r+s} < 2E_{r+s}, \quad \sigma(a - c) \leq E_{r+s} < 2E_{r+s}.$$

By the Fineness Condition and (4.2),

$$(2E_{r+s})^2 < R^{2(r+s)} \leq Q(a).$$

Lemma 3.5, applied with $\varepsilon \equiv 2E_{r+s}$, therefore shows that the origin is not a nonnegative rational affine combination of (a, b, c) . Hence every corresponding image three-cycle has winding number 0 by Theorem 4.6(2). Summing over all image three-cycles and cancelling the contribution of interior edges gives

$$0 = W(z_0^s \cdot P_r) + W(z_0^r \cdot P_s) - W(P_{r+s}) = I_r + I_s - I_{r+s}.$$

The last equality follows from $z_0 = R > 0$ and Remark 4.5. $\square$

Theorem 5.5 (Winding number of the power cycle). *If $4E_n^2 < R^{2n}$, then $I_n = n$.*

Proof. Suppose that $4E_n^2 < R^{2n}$. We first verify the Fineness Condition for every $1 \leq r < n$. Since $h = 2R/M$, for any $k \geq 1$ we have

$$E_k = k(2R)^{k-1}h = \frac{k(2R)^k}{M},$$

and hence

$$q_k \equiv \frac{4E_k^2}{R^{2k}} = \frac{4k^2 4^k}{M^2}.$$

Moreover,

$$\frac{q_{k+1}}{q_k} = 4 \left(\frac{k+1}{k} \right)^2 > 1,$$

so $(q_k)_{k \geq 1}$ is strictly increasing. Hence $q_n < 1$ implies $q_r < q_n < 1$, or equivalently $4E_r^2 < R^{2r}$, for every $1 \leq r < n$.

We now prove the result by induction on n . For $n = 1$, Lemma 5.3 gives $I_1 = 1$. For $n \geq 2$, the induction hypothesis and the preceding Fineness Condition give $I_{n-1} = n - 1$. Apply Theorem 5.4 with $r = n - 1$ and $s = 1$. Its required Fineness Condition is the assumed condition for n , and consequently

$$I_n = I_{n-1} + I_1 = n,$$

where Lemma 5.3 was used in the last equality. $\square$

Remark 5.6 (On the choice of the starting vertex z_0). In Definition 5.1 we fixed $z_0 = R$ (the rightmost point on the boundary of the square) for concreteness. This choice is immaterial: the winding number $W(P)$ of any closed vertex list is invariant under cyclic permutation of the list. Indeed, the sum in $W(P) = \sum_i \chi(p_i, p_{i+1})$ is taken over all oriented edges exactly once; a cyclic shift merely reorders the summands, so the total is unchanged. Therefore, any other boundary vertex z_k ($0 \leq k < 4M$) could serve as the starting point, and the resulting power cycle would have the same winding number I_n . The half-open convention ensures consistent treatment of the edge that wraps around the chosen starting point, so no special handling is required.

Remark 5.7 (Proof of scale-invariance via triangulation). Fix two consecutive boundary vertices z_i, z_{i+1} and assume $4E_{n+1}^2 < R^{2n+2}$. For $0 \leq j \leq 4M$, set

$$A_j \equiv z_i z_j^n, \quad B_j \equiv z_{i+1} z_j^n.$$

For each $0 \leq j < 4M$, split the four-point list

$$(A_j, B_j, B_{j+1}, A_{j+1}) = (z_i z_j^n, z_{i+1} z_j^n, z_{i+1} z_{j+1}^n, z_i z_{j+1}^n)$$

along $A_j B_{j+1}$ into

$$(A_j, B_j, B_{j+1}), \quad (A_j, B_{j+1}, A_{j+1}).$$

By Lemmas 3.3 and 3.2, the “horizontal” and “vertical” changes in this four-point list are bounded respectively by $h(2R)^n$ and $n(2R)^n h$. Their sum is

$$E_{n+1} = (n+1)(2R)^n h.$$

Thus

$$\sigma(U - A_j) \leq E_{n+1} < 2E_{n+1} \quad \text{for } U \in \{B_j, B_{j+1}, A_{j+1}\}.$$

Since

$$Q(A_j) = Q(z_i)Q(z_j)^n \geq R^{2(n+1)} > (2E_{n+1})^2,$$

Lemma 3.5, with $\varepsilon \equiv 2E_{n+1}$, excludes the origin from both three-point lists. Their three-cycles therefore have winding number 0 by Theorem 4.6(2). Summing over j cancels all diagonal and interior edges, leaving

$$0 = W(z_{i+1} \cdot P_n) - W(z_i \cdot P_n),$$

so $W(z_i \cdot P_n) = W(z_{i+1} \cdot P_n)$.

Theorem 5.8 (Conditional scale invariance). *Let $a \in K^\times$, $R > 0$, $M \geq 1$, $n \geq 1$, and assume the fineness condition $4E_{n+1}^2 < R^{2n+2}$. Then*

$$W(a \cdot P_n) = I_n.$$

Proof. Write $a = x + iy$ with $x, y \in \mathbb{Q}$, and set

$$t \equiv \frac{R}{\max\{|x|, |y|\}}, \quad \tilde{a} \equiv ta.$$

Since $a \neq 0$, we have $t \in \mathbb{Q}$ and $t > 0$, and

$$\max\{|\operatorname{Re}(\tilde{a})|, |\operatorname{Im}(\tilde{a})|\} = R.$$

Thus $\tilde{a}$ is the intersection of the ray from 0 through a with $\partial\operatorname{Sq}(R)$. Hence Remark 4.5 gives

$$W(a \cdot P_n) = W\left(\frac{1}{t}\tilde{a} \cdot P_n\right) = W(\tilde{a} \cdot P_n).$$

Insert $\tilde{a}$ into the boundary subdivision if necessary. This only shortens one boundary segment, so the triangulation argument of Remark 5.7 remains valid and yields

$$W(\tilde{a} \cdot P_n) = W(P_n) = I_n.$$

The result follows. □

6 The existence of suitable radius and mesh resolution

Throughout, $f(X) = \sum_{j=0}^n a_j X^j \in K[X]$ with $n \geq 1$, $a_n \neq 0$, and a rational $\varepsilon > 0$. Let $A_j \equiv \sigma(a_j)$ and

$$C \equiv \sum_{j=0}^{n-1} 2^j A_j, \quad G \equiv CR^{n-1}, \quad m \equiv Q(a_n) R^{2n}, \quad L \equiv \sum_{j=1}^n j A_j (2R)^{j-1}.$$

Lemma 6.1 (Choice of R). *There is an integer $R \geq 1$ such that*

$$Q(a_n) R^2 > 2C^2. \tag{5.1}$$

For any such R one also has $2G^2 < m$.

Proof. One can choose a sufficiently large integer R such that $R^2 > 2C^2/Q(a_n)$. Multiplying (5.1) by the positive factor R^{2n-2} gives $2G^2 < m$. □

Lemma 6.2 (Choice of M). *Let $R \geq 1$ be an integer satisfying (5.1), whose existence is guaranteed by Lemma 6.1. Then there exists an effectively found integer $M \geq 1$ such that, writing $h \equiv 2R/M$ and $D \equiv A_n E_n$, all three strict rational number inequalities hold:*

$$2hL < \varepsilon, \quad (D + G)^2 < m, \quad 4E_{n+1}^2 < R^{2n+2}. \tag{5.2}$$

The last inequality also implies $4E_n^2 < R^{2n}$.

Proof. For $k = n, n + 1$, write

$$e_k \equiv 2Rk(2R)^{k-1},$$

and set $d_0 \equiv A_n e_n$. Thus $E_k = e_k/M$ for $k = n, n + 1$, and $D = d_0/M$. By Lemma 6.1, $\delta \equiv m - G^2 > 0$. Choose M large enough such that

$$\frac{4RL}{M} < \varepsilon, \quad \frac{2Gd_0}{M} < \frac{\delta}{2}, \quad \frac{d_0^2}{M^2} < \frac{\delta}{2}, \quad \frac{4e_{n+1}^2}{M^2} < R^{2n+2}.$$

Then the first is exactly $2hL < \varepsilon$, the fourth gives $4E_{n+1}^2 < R^{2n+2}$, and

$$(D + G)^2 = G^2 + \frac{2Gd_0}{M} + \frac{d_0^2}{M^2} < G^2 + \frac{\delta}{2} + \frac{\delta}{2} = m.$$

Finally, the proof of Theorem 5.5 shows that

$$q_k \equiv \frac{4E_k^2}{R^{2k}}$$

is strictly increasing in k . Hence $q_{n+1} < 1$ implies $q_n < 1$, which is precisely $4E_n^2 < R^{2n}$. $\square$

7 The existence of a good grid vertex

Choose R, M, h as in §6; in particular, $4E_{n+1}^2 < R^{2n+2}$. Consider the finite rational grid

$$\mathcal{G} \equiv \{-R + ih + i(-R + jh) : 0 \leq i, j \leq M\},$$

listed lexicographically (the i -coordinate outer, j inner); it has exactly $(M + 1)^2$ points. As in §5, its boundary $\partial\mathcal{G} = \partial\text{Sq}(R)$ is subdivided into $4M$ segments, each of quadrance h^2 . Each of the M^2 cells is split by its positive diagonal into two triangles.

Theorem 7.1 (Existence of a good grid vertex). *There exists $z \in \mathcal{G}$ with*

$$Q(f(z)) < \varepsilon^2.$$

Proof. Suppose, for contradiction, that

$$Q(f(z)) \geq \varepsilon^2 \quad \text{for all } z \in \mathcal{G}. \quad (6.1)$$

(a) *Image three-point lists exclude 0.* For two vertices z, w of one grid triangle, $\sigma(w - z) \leq 2h$; since all grid points lie in $\mathcal{G}$, each grid vertex has real and imaginary parts bounded by R in absolute value, so for vertices z, w we have $|\Re(z)|, |\Im(z)|, |\Re(w)|, |\Im(w)| \leq R$. Lemma 3.4 and (5.2) give, for vertices z, w of the same triangle,

$$\sigma(f(w) - f(z)) \leq \sigma(w - z) \sum_{j=1}^n j A_j (2R)^{j-1} = \sigma(w - z) L \leq 2hL < \varepsilon. \quad (6.2)$$

Let $p_0 \equiv f(z_0), p_1 \equiv f(z_1), p_2 \equiv f(z_2)$ be the three image values of a grid triangle. If the origin were a nonnegative rational affine combination of (p_0, p_1, p_2) , there would exist nonnegative rationals $\theta_0, \theta_1, \theta_2$ with $\theta_0 + \theta_1 + \theta_2 = 1$ and $0 = \theta_0 p_0 + \theta_1 p_1 + \theta_2 p_2$. Then

$$p_0 = \theta_1(p_0 - p_1) + \theta_2(p_0 - p_2),$$

so by (6.2) and the affine-exclusion argument (Lemma 3.5) $\sigma(p_0) < \varepsilon$, hence $Q(p_0) \leq \sigma(p_0)^2 < \varepsilon^2$, contradicting (6.1). Thus no image three-point list admits such an affine combination

of the origin, and by Theorem 4.6(2) each image three-cycle has winding number 0: $W(f(z_0), f(z_1), f(z_2), f(z_0)) = 0$.

(b) *Edge cancellation.* Summing the winding numbers of all $2M^2$ image three-cycles: every interior edge of the triangulation occurs twice, in opposite orientations, and cancels by Theorem 4.6(1). What remains is the image of the outer boundary:

$$W(f(\partial\mathcal{G})) = W(f((z_j)_{j=0,1,2,\dots,4M})) = 0. \quad (6.3)$$

(c) *The boundary winding number is the leading-term winding number.* Write $f(z) = l(z) + g(z)$ with $l(z) \equiv a_n z^n$ and $g(z) \equiv \sum_{j < n} a_j z^j$. At a boundary grid vertex, $\sigma(z) \leq 2R$, $Q(z) \geq R^2$, and $R \geq 1$; hence

$$\sigma(g(z)) \leq \sum_{j < n} A_j (2R)^j \leq CR^{n-1} = G, \quad Q(l(z)) \geq Q(a_n) R^{2n} = m. \quad (6.4)$$

For consecutive boundary vertices z, z' (with $\sigma(z' - z) = h$), the power estimate gives

$$\sigma(l(z') - l(z)) \leq D. \quad (6.5)$$

Join the leading-term closed vertex list $(l(z_j))_j$ to the full-value closed vertex list $(l(z_j) + g(z_j))_j$ by the closed four-vertex list $(l(z_j), f(z_j), f(z_{j+1}), l(z_{j+1}), l(z_j))$, and split this list along a diagonal into the two closed three-cycles

$$(l(z_j), f(z_j), f(z_{j+1}), l(z_j)), \quad (l(z_j), f(z_{j+1}), l(z_{j+1}), l(z_j)).$$

Every vertex u of either three-cycle satisfies $\sigma(u - l(z_j)) \leq D + G$ (using (6.4)–(6.5)). If the origin were a nonnegative rational affine combination of either determining three-point list, the same argument as in Lemma 3.5 would give $Q(l(z_j)) \leq (D + G)^2 < m$, contradicting (6.4). Since neither list admits such a combination, each three-cycle has winding number 0, again finite edge cancellation yields

$$W(f(\partial\mathcal{G})) = W((a_n z_j^n)_{j=0,1,2,\dots,4M}). \quad (6.6)$$

(d) *The boundary winding number is n .* Lemma 6.2 also gives the degree- n Fineness Condition. Therefore, by Theorems 5.5 and 5.8,

$$W((a_n z_j^n)_{j=0,1,2,\dots,4M}) = W(a_n \cdot P_n) = I_n = n. \quad (6.7)$$

Equations (6.3) and (6.7) give $0 = n$, contradicting $n \geq 1$. Hence (6.1) is false, i.e. some grid vertex satisfies $Q(f(z)) < \varepsilon^2$. $\square$

8 Proof of main theorem

Theorem 8.1 (Main theorem). *For any nonconstant polynomial f with rational complex number coefficients and every rational number $\varepsilon > 0$, there exists a rational complex number z such that*

$$Q(f(z)) < \varepsilon^2.$$

Proof. Write $f(X) = \sum_{j=0}^n a_j X^j$ with $n \geq 1$, $a_n \neq 0$. Let $A_j \equiv \sigma(a_j)$ and $\nu \equiv Q(a_n) > 0$. By Lemmas 6.1 and 6.2 there are integers $R \geq 1$, $M \geq 1$ satisfying the three strict inequalities (5.2). Form the finite grid $\mathcal{G}$ and apply Theorem 7.1 : there exists some $z \in \mathcal{G}$ satisfying $Q(f(z)) < \varepsilon^2$. $\square$

The proof of theorem is also constructive with an algorithm for finding an approximate zero: construct the finite grid $\mathcal{G}$ used in its proof and test its vertices one by one until finding a vertex z such that $Q(f(z)) < \varepsilon^2$. The theorem guarantees that the algorithm will terminate after finitely many steps. This exhaustive search is generally slow; finding an elementary proof with more effective algorithm is left for future work.

Remark 8.2. Given a positive integer k , let k_1 be any integer satisfying $\frac{1}{k_1^2} < \frac{1}{k}$. Let $\varepsilon \equiv \frac{1}{k_1}$. By the main theorem (Theorem 8.1), there exists a rational complex number z such that

$$Q(f(z)) < \frac{1}{k_1^2} < \frac{1}{k}.$$

This is one of Wildberger's formulation of the Fundamental Theorem of Algebra over Rationals, which he named FTA I.

9 Complete approximate factorization over Rationals

We now establish a complete approximate factorization theorem: every nonconstant polynomial over K can be approximated, up to any prescribed precision, by a monic (up to its leading coefficient) product of linear factors over K .

9.1 Polynomial coefficient quadrance

For a polynomial

$$P(X) = \sum_{j=0}^m c_j X^j \in K[X],$$

define its *coefficient quadrance* by

$$Q_{\text{coeff}}(P) \equiv \max_{0 \leq j \leq m} Q(c_j).$$

For $P, H \in K[X]$, the elementary inequality $Q(u + v) \leq 2(Q(u) + Q(v))$ gives

$$Q_{\text{coeff}}(P + H) \leq 2(Q_{\text{coeff}}(P) + Q_{\text{coeff}}(H))$$

and, when multiplied by a linear factor,

$$Q_{\text{coeff}}((X - z)P) \leq 2(1 + Q(z)) Q_{\text{coeff}}(P) \quad (z \in K),$$

because the coefficients of $(X - z)P$ are of the form $c_{j-1} - z c_j$, whose quadrance is bounded by $2(Q(c_{j-1}) + Q(z)Q(c_j))$.

The main goal of this section is to prove the following theorem.

Theorem 9.1 (Complete approximate factorization over Rationals). *Let $f \in K[X]$ be nonconstant with leading coefficient $a_n \neq 0$ and degree $n \geq 1$. For any rational $\varepsilon > 0$ there exist $z_1, \dots, z_n \in K$ such that*

$$Q_{\text{coeff}}\left(f - a_n \prod_{i=1}^n (X - z_i)\right) < \varepsilon.$$

9.2 Proof of Theorem 9.1

We prove the theorem by induction on $n \geq 1$.

Base case $n = 1$. Write $f(X) = a_1 X + a_0$ with $a_1 \neq 0$. Take $z_1 = -a_0/a_1 \in K$. Then

$$f(X) - a_1(X - z_1) = 0,$$

so the desired inequality holds with any $\varepsilon > 0$.

Inductive step. Assume the theorem holds for all polynomials of degree $< n$ (with $n \geq 2$). Let f be as above and fix $\varepsilon > 0$.

Set

$$\eta \equiv \frac{\min(1, \varepsilon)}{4},$$

so that $\eta^2 < \varepsilon/4$. Apply the main theorem (Theorem 8.1) to f with precision η : there exists $z_1 \in K$ such that

$$Q(f(z_1)) < \eta^2 < \varepsilon/4.$$

Let T be a large integer satisfying

$$Q(z_1) \leq T.$$

Let $r \equiv f(z_1)$. By our assumption,

$$Q(r) < \frac{\varepsilon}{4} \quad (8.1)$$

. By polynomial division by the monic linear factor $X - z_1$, we obtain

$$f(X) = (X - z_1)g(X) + r,$$

where $g \in K[X]$ has degree $n - 1$ and leading coefficient a_n . Set

$$\varepsilon_1 \equiv \frac{\varepsilon}{8(T + 1)}.$$

Since $n - 1 \geq 1$ (because $n \geq 2$), the induction hypothesis applies to g : there exist $z_2, \dots, z_n \in K$ such that

$$Q_{\text{coeff}}\left(g - a_n \prod_{i=2}^n (X - z_i)\right) < \varepsilon_1. \quad (8.2)$$

Define the remainder polynomial

$$h(X) \equiv g(X) - a_n \prod_{i=2}^n (X - z_i) = \sum_{j=0}^{n-1} b_j X^j,$$

thus $Q(b_j) < \varepsilon_1$ for every $j = 0, 1, \dots, n - 1$.

Now compute the global error:

$$\begin{aligned} f(X) - a_n \prod_{i=1}^n (X - z_i) &= (X - z_1)g(X) + r - (X - z_1)a_n \prod_{i=2}^n (X - z_i) \\ &= (X - z_1)h(X) + r. \end{aligned}$$

Let

$$(X - z_1)h(X) = \sum_{j=0}^n c_j X^j,$$

where

$$c_0 = -z_1 b_0, \quad c_j = b_{j-1} - z_1 b_j \quad (1 \leq j \leq n - 1), \quad c_n = b_{n-1}.$$

Thus the coefficients of the polynomial $f(X) - a_n \prod_{i=1}^n (X - z_i)$ are

$$c_0 + r, \quad c_1, \quad \dots, \quad c_{n-1}, \quad c_n.$$

We next estimate their quadrances. For $1 \leq j \leq n - 1$,

$$Q(c_j) \leq 2(Q(b_{j-1}) + Q(z_1)Q(b_j)) < 2(1 + T)\varepsilon_1 = \frac{\varepsilon}{4} < \varepsilon.$$

For the leading coefficient,

$$Q(c_n) = Q(b_{n-1}) < \varepsilon_1 < \varepsilon.$$

For the constant term,

$$Q(c_0 + r) \leq 2(Q(r) + Q(z_1)Q(b_0)) < \frac{\varepsilon}{2} + 2T\varepsilon_1 = \frac{\varepsilon}{2} + \frac{T\varepsilon}{4(T+1)} < \frac{3\varepsilon}{4} < \varepsilon,$$

by using (8.1) and $Q(b_0) < \varepsilon_1$.

Therefore every coefficient of the error polynomial has quadrance strictly less than ε , i.e.

$$Q_{\text{coeff}}\left(f - a_n \prod_{i=1}^n (X - z_i)\right) < \varepsilon.$$

This completes the induction and proves the theorem.

Proposition 9.2 (Wildberger’s FTA II). *Let $f \in K[X]$ be nonconstant with leading coefficient $a_n \neq 0$ and degree $n \geq 1$. For any integer $k \geq 1$, there exist $z_1, \dots, z_n \in K$ such that, if*

$$f(X) - a_n \prod_{i=1}^n (X - z_i) = \sum_{j=0}^n e_j X^j,$$

then

$$\sum_{j=0}^n Q(e_j) < \frac{1}{k}.$$

Proof. Apply Theorem 9.1 with $\varepsilon = 1/(k(n+1))$. Then $Q(e_j) < \varepsilon$ for every j , and hence

$$\sum_{j=0}^n Q(e_j) < (n+1)\varepsilon = \frac{1}{k}.$$

□

Remark 9.3. Theorem 8.1 and Proposition 9.2 are, respectively, Wildberger’s two formulations of the Fundamental Theorem of Algebra over Rationals.

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